Last name _____

First name _____

LARSON—MATH 310–CLASSROOM WORKSHEET 23
Pseudo-Inverses and Least Squares.

Review Given square A with linearly independent columns...

1. We know $A = QR$ for an orthogonal matrix Q and an upper-triangular matrix R. Find Q.

2. We know orthogonal matrices Q are invertible with inverse $Q^{-1} = Q^T$. So, $Q^T A = R$. Use this fact to find R (In particular $R_{i,j} = (Q^T A)_{i,j}$. So $R_{i,j}$ is the inner product of $\vec{q}_i$ and $\vec{a}_j$).

3. Check that A has linearly independent columns (for a square matrix that's enough to guarantee that A has an inverse A^{-1}).

4. We know $A^{-1} = (QR)^{-1} = R^{-1}Q^{-1} = R^{-1}Q^T$. So to find A^{-1} we just need R^{-1}. Since R is upper-triangular with no zero entry on its diagonal we know it has an inverse. Solve
$$R\left[\vec{x}_1\vec{x}_2\right] = \left[\vec{e}_1\vec{e}_2\right],$$
by solving $R\vec{x}_1 = \vec{e}_1$ and $R\vec{x}_2 = \vec{e}_2$. (These correspond to systems that can be backsolved.)

5. Let $R^{-1} = \left[\vec{x}_1\vec{x}_2\right]$. Check that $RR^{-1} = I$.

6. Now find A^{-1} (we know $A^{-1} = R^{-1}Q^{-1} = R^{-1}Q^T$).

7. Check that $AA^{-1} = I$.

Matrix Pseudo Inverses!

1. Show: if A is a matrix with linearly independent columns then $A^T A$ is invertible.

We showed: if A is a matrix with linearly independent columns then $(A^T A)^{-1}A^T$ is the left-inverse of A (called a *pseudo-inverse*).

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

$$(A^T A)^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

2. Find the (left) pseudo-inverse of A. Call it $A^\dagger$.

   Since A is square, $A^\dagger$ must be the true inverse of A.

3. Check that $A^\dagger A = I$.

4. Let

$$A = \begin{bmatrix} 2 & 0 \\ -1 & 1 \\ 0 & 2 \end{bmatrix}$$

$$\vec{b} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

5. Explain why the equation $A\vec{x} = \vec{b}$ has no exact solution.

6. Consider and test some *approximate* solutions $\vec{x}$ to $A\vec{x} = \vec{b}$.

7. Prove: that $\tilde{x} = (A^T A)^{-1} A^T \vec{b}$ is the "best" approximate solution to $A\vec{x} = \vec{b}$ (in the sense that $||A\tilde{x} - \vec{b}|| \leq ||A\vec{x} - \vec{b}||$ for every vector $\vec{x}$).